

Project Design Phase

Problem – Solution Fit

Date	01-10-2026
Team ID	SWTID-2026-8759
Project Name	Import Data using Transform Maps (Spreadsheet)
Maximum Marks	2 Marks

Problem – Solution Fit Template

The problem is that bulk employee data arrives in spreadsheets and is slow and error-prone to enter into ServiceNow by hand. The solution, Import Sets with a Transform Map and Coalesce, loads that data accurately, avoids duplicates on repeat loads, and shows the result in reports and a dashboard.

Purpose

- Solve the bulk data entry problem in a way that fits how employee data is already received (Excel / Google Sheets)
- Increase adoption by building on tools users already use: spreadsheets and the ServiceNow Load Data screen
- Build trust by solving a frequent annoyance: repeated manual entry and duplicate records
- Understand the existing manual import situation in order to improve it for admins and HR teams

Canvas

Canvas Section	Content
1. Customer Segment(s) (CS)	ServiceNow admins, HR teams and business analysts in IT services companies who receive employee data in Excel and

	must load it into ServiceNow. Includes beginners learning the data import process
6. Customer Constraints (CC)	Little or no coding skill; data comes from outside as spreadsheets; limited time; needs admin access to create tables and maps; data must stay accurate with no duplicates
5. Available Solutions (AS)	Manual record-by-record entry (slow, error-prone). Custom scripts or integrations (need developer skills). Plain file import without field mapping or duplicate control (data lands in wrong fields or repeats)
2. Jobs-to-be-Done / Problems (J&P)	Move bulk employee records from Excel into ServiceNow quickly and correctly. Keep existing records up to date when names or emails change. Get department and location insights from the data
9. Problem Root Cause (RC)	Employee data sits in spreadsheets with different structure from ServiceNow tables. Manual entry is repetitive. Without a unique key check, repeated imports create duplicate records
7. Behaviour (BE)	Maintain employee lists in Excel or Google Sheets, share them by email, copy and paste into forms, and check for duplicates by eye
3. Triggers (TR)	A new batch of employees is onboarded. An updated spreadsheet arrives with changed names or emails. A manager asks for headcount by department or location
4. Emotions: Before / After (EM)	Before: stressed about errors and duplicates, frustrated by repeated work. After: confident, in control, and trusting the data

10. Your Solution (SL)

Load the .xlsx into a staging table with Import Sets, map fields to the Employee Test table with a Transform Map, enable Coalesce on Employee ID so repeat imports update instead of duplicate, then show the data in three reports on the Employee Analytics Dashboard. Tested result: 15 records inserted first, then 2 inserts and 2 updates, and 4 ignored on an identical re-import

8.1 Channels of Behaviour: Online (CH)

Google Sheets / Excel files, email sharing, ServiceNow Load Data and Transform Map screens, dashboards in the browser

8.2 Channels of Behaviour: Offline (CH)

Spreadsheets saved on local systems, printed employee lists, verbal or in-person update requests from managers

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Project Design Phase: Solution Architecture

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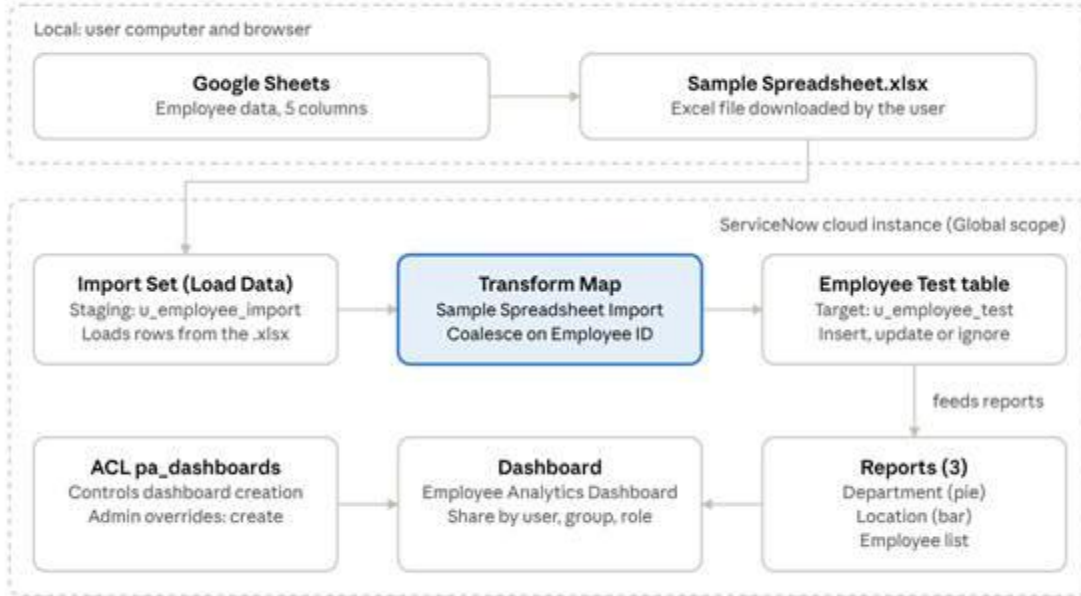
Solution Architecture

The solution bridges the business problem (bulk employee data arriving in Excel and being entered manually) with a ServiceNow low-code solution. Its goals are to:

- Find the best tech solution: Import Sets and Transform Maps load spreadsheet data accurately without scripting
- Describe the structure and behavior of the solution to project stakeholders through the diagram and data flow below
- Define features, phases and requirements: data preparation, target table, import, transform, coalesce, reports and dashboard
- Provide specifications for how the solution is defined, managed and delivered: tables, field maps, reports and access controls

Solution Architecture Diagram

Coalesce in the Transform Map decides insert, update or ignore



solution architecture · 6 components, 2 zones

The file is staged, then the Transform Map applies Coalesce on Employee ID to insert, update or ignore each row before the reports read the Employee Test table.

Data Flow

1. User prepares employee data in Google Sheets and downloads it as Sample Spreadsheet.xlsx
2. Load Data uploads the file into the staging table Employee Import (u_employee_import) and creates an import set
3. Transform Map Sample Spreadsheet Import maps source fields to the target table Employee Test (u_employee_test)
4. Coalesce on Employee ID decides whether each row is inserted as a new record, updates an existing record, or is ignored when nothing changed
5. Reports (Employees by Department, Employees by Location, Employee List Report) read from Employee Test
6. The reports are shown together on the Employee Analytics Dashboard, shared to users, groups and roles as needed

Key Components

Component	Role
Google Sheets / .xlsx file	Source of employee data (Employee ID, Name, Email, Department, Location)
Import Set table (u_employee_import)	Staging area for loaded rows
Transform Map	Maps source fields to target fields and applies Coalesce on Employee ID
Employee Test table (u_employee_test)	Target table holding the final employee records
Reports and Dashboard	Department pie chart, Location bar chart, employee list, all on one dashboard
ACL pa_dashboards	Controls dashboard creation (Admin overrides enabled for the create operation)